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JAVA PROJECT REPORT ON

RESTAURANT ORDER PROCESSING BILLING

Submitted by

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DECLARATION

I hereby declare that the project entitled “RESTAURANT ORDER PROCESSING BILLING” submitted in partial fulfillment of the requirements for the course JAVA PROGRAMMING is a **record of my original work** carried out under the guidance of **MR.DHARANEESH**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

___JEYASINGH.T_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"RESTAURANT ORDER PROCESSING BILLING"** is a **bona fide work** carried out by **JEYASINGH.T B.Sc/2025-2028**, [COMPUTER SCIENCE], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: KGCAS

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on

____10/04/2026_____

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ABSTRACT

The **Restaurant Order Processing and Billing System** is a Java-based application developed to automate the process of taking orders and generating bills in restaurants. In many small restaurants, billing is done manually, which can lead to calculation errors, delays, and inefficient service.

This project provides a simple and effective solution by allowing users to select food items from a predefined menu, enter the quantity, and automatically calculate the total bill along with applicable tax. The system generates a clear and structured invoice that includes item details, subtotal, tax, and final amount.

The application is developed using Core Java concepts such as classes, objects, loops, and conditional statements. It can be implemented as a console-based program and is easy to use.

The main objective of this system is to improve billing accuracy, reduce manual effort, and provide faster service to customers. This project also demonstrates the practical use of Java programming in developing real-world applications.

1. INTRODUCTION

In restaurants, billing is an important task that directly affects customer satisfaction. Traditional manual billing methods are slow and prone to errors. Staff may miscalculate totals or delay service during busy hours.

With the advancement of technology, automated billing systems have become essential. The Restaurant Billing System is designed to simplify order processing and bill generation.

This system enables users to select menu items, enter quantities, and generate bills automatically. It reduces human effort and ensures accuracy.

1.1 Objective

To develop a Java-based billing system

- To automate order processing
- To reduce manual errors
- To generate accurate bills
- To improve service efficiency

1.2 Overview

The system is a menu-driven Java application. It displays available food items and allows the user to select items and enter quantities. The system calculates the total amount and adds tax automatically.

1.3 Features

Menu display

- Order processing
- Automatic bill calculation
- Tax calculation
- Invoice generation
- Easy to use interface

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Manual billing systems have several issues:

- Incorrect calculations
- Slow service
- Difficulty in managing orders
- Lack of proper records

This system solves these problems by automating billing

2.2 Project Overview

The system:

- Displays menu items
- Accepts user input
- Calculates total cost
- Generates final bill

2.3 Scopes and Limitations

Scope:

- Restaurants
- Cafeterias
- Small food businesses

Limitations:

- No database integration
- Console-based system
- Limited features

3. SYSTEM DESIGN

System Design is the phase in software development where the overall structure and architecture of the system are planned before implementation. It involves dividing the

system into smaller modules, defining their functions, and designing how these modules interact with each other to achieve the desired output.

In the **Restaurant Order Processing and Billing System**, system design focuses on creating a simple and efficient structure that allows users to select food items, enter quantities, and generate bills automatically. The system is designed as a menu-driven application to ensure ease of use and smooth operation

3.1 Modules

1. Menu Display Module

This module is responsible for displaying the list of available food items along with their prices.

- Shows menu items to the user
- Helps the user choose items easily
- Displays data in a clear format

Example:

- Burger – ₹100
- Pizza – ₹200
- Juice – ₹50

---

2. Order Processing Module

This module handles the selection of food items and quantity input from the user.

- Accepts user choice
- Accepts quantity of items
- Stores order details temporarily
- Allows multiple item selection

This module ensures proper input handling and smooth order processing.

---

3. Billing Module

This module performs all the calculations related to the bill.

- Calculates total price of items
- Computes subtotal
- Calculates tax (GST – 5%)
- Generates final bill amount

Formula Used:

- Total = Price × Quantity
- Tax = 5% of Total
- Final Bill = Total + Tax

---

4. Invoice Generation Module

This module generates the final bill in a structured format.

- Displays item name, quantity, and price
- Shows subtotal and tax
- Displays final payable amount
- Produces a clear and readable bill

Example Output:

Item	Qty	Price
Burger	2	200
Juice	1	50
Subtotal		250
Tax (5%)		12.5
Total		262.5

---

5. Exit Module

This module is used to terminate the program safely.

- Allows the user to exit the system
- Ends program execution properly

---

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Hotel & Restaurant Billing System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the existing manual billing system was studied and analyzed.

Identified Problems:

- Manual calculations may lead to errors
- Billing process is time-consuming
- Difficult to manage multiple orders
- No proper record maintenance
- Delays in customer service

Based on these issues, the need for an automated billing system was identified.

4.2. System Design

In this stage, the structure of the system was planned.

- Divided system into modules:
    - Menu Module
    - Order Module
    - Billing Module
    - Invoice Module

    - Exit Module

- Designed a menu-driven interface
- Planned input/output flow

This helps in making the system organized and easy to develop.

4.3. Program Development

Technologies Used:

- Java (JDK 8 or above)
- IDE (Eclipse / IntelliJ / VS Code)

Concepts Used:

- Classes and Objects
- Loops (for menu repetition)
- Conditional statements (switch-case)
- Methods (for modular coding)

Development Steps:

- Created main class
- Implemented menu display
- Added user input handling
- Developed billing logic
- Generated final output

4.4. Testing and Validation

Testing Done:

- Checked correct menu display
- Verified user input handling
- Tested billing calculations
- Validated tax calculation
- Checked invalid input handling

Sample Test Cases:

Test Case	Expected Result
Select item	Correct item displayed
Enter quantity	Correct value accepted
Calculate bill	Accurate total generated
Invalid input	Error handled properly

4.5. Result and Documentation

After successful testing, the results were recorded.

Results:

- System generates accurate bills
- Reduces manual effort

- Works efficiently

Documentation was prepared to explain:

- System design
- Modules
- Code
- Output

4.6. Algorithm

Step-by-step process:

1. Start the program
2. Display menu
3. Ask user to select item
4. Enter quantity
5. Calculate item total
6. Add to overall total
7. Repeat for multiple items
8. Calculate tax (5%)
9. Calculate final bill
10. Display bill
11. End program

4.7. Flowchart

Start

↓

Display Menu

↓

User selects item

↓

Enter quantity

↓

Add to total

↓

Repeat until user enters 0

↓

Generate Bill

↓

Display Total + Tax

↓

End

5. SOURCE CODE

```
import java.util.*;


class RestaurantSystem {
    static Scanner sc = new Scanner(System.in);


    static String[] items = {"C.BRIYANI", "M.BRIYANI", "MANDHI", "GRILLED
CHICKEN", "COKE"};
    static double[] prices = {150, 350, 900, 300, 50};
```

```
public static void main(String[] args) {
    int choice;
    int[] quantity = new int[items.length];
    double total = 0;

    System.out.println("===== Welcome to ABU DHABI RESTAURANT =====");

    do {
        displayMenu();
        System.out.print("Enter item number to order: ");
        choice = sc.nextInt();

        if (choice >= 1 && choice <= items.length) {
            System.out.print("Enter quantity: ");
            int qty = sc.nextInt();
            quantity[choice - 1] += qty;
            total += prices[choice - 1] * qty;
        } else if (choice != 0) {
            System.out.println("Invalid choice!");
        }

    } while (choice != 0);

    generateBill(quantity, total);
}

static void displayMenu() {
    System.out.println("\nMenu:");
    for (int i = 0; i < items.length; i++) {
```

```
            System.out.println((i + 1) + ". " + items[i] + " - Rs." + prices[i]);
        }
        System.out.println("0. Finish Order");
    }

    static void generateBill(int[] quantity, double total) {
        System.out.println("\n===== BILL =====");
        for (int i = 0; i < items.length; i++) {
            if (quantity[i] > 0) {
                System.out.println(items[i] + " x " + quantity[i] + " = Rs." + (prices[i] *
quantity[i]));
            }
        }

        double tax = total * 0.05;
        double grandTotal = total + tax;

        System.out.println("-----------------------");
        System.out.println("Total: Rs." + total);
        System.out.println("Tax (5%): Rs." + tax);
        System.out.println("Grand Total: Rs." + grandTotal);
        System.out.println("========================");
        System.out.println("Thank You! Visit Again!");
    }
}
```

6. INPUT AND OUTPUT SCREENS

```
===== Welcome to ABU DHABI RESTAURANT =====

Menu:
1. C.BRIYANI - Rs.150.0
2. M.BRIYANI - Rs.350.0
3. MANDHI - Rs.900.0
4. GRILLED CHICKEN - Rs.300.0
5. COKE - Rs.50.0
0. Finish Order
Enter item number to order: 1
Enter quantity: 1

Menu:
1. C.BRIYANI - Rs.150.0
2. M.BRIYANI - Rs.350.0
3. MANDHI - Rs.900.0
4. GRILLED CHICKEN - Rs.300.0
5. COKE - Rs.50.0
0. Finish Order
Enter item number to order: 0

===== BILL =====
C.BRIYANI x 1 = Rs.150.0
-----------------------
Total: Rs.150.0
Tax (5%): Rs.7.5
Grand Total: Rs.157.5
=======================
Thank You! Visit Again!
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Restaurant Order Processing and Billing System** was developed to automate the billing process in restaurants. The system allows users to select food items, enter quantities, and generate bills automatically in a simple and efficient manner.

During the development of this project, various Core Java concepts such as classes, objects, loops, and conditional statements were applied. The system was tested with different inputs to ensure that it performs calculations accurately and handles user input correctly.

The results show that the system effectively reduces manual effort and minimizes errors in billing. It also improves the speed of service, especially during busy hours. The user-friendly design makes it easy for restaurant staff to operate the system without difficulty.

7.2 Conclusion

The **Restaurant Order Processing and Billing System** successfully demonstrates the practical application of Java programming in developing a real-world solution.

The system automates the billing process, reduces calculation errors, and provides faster and more efficient service to customers. It simplifies restaurant operations by generating accurate and well-structured invoices.

This project helps in understanding how programming concepts can be applied to solve real-life problems. Overall, the system meets its objectives and serves as a strong foundation for developing more advanced restaurant management applications in the future.

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